

Linear Data Structures: Linked List

لنکڈ لسٹ زنجیر کی طرح نوڈز پر مشتمل ہے جو میموری میں بکھرے ہو سکتے ہیں۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 16 of 24

Linear Data Structures: Linked List

LEARN IT

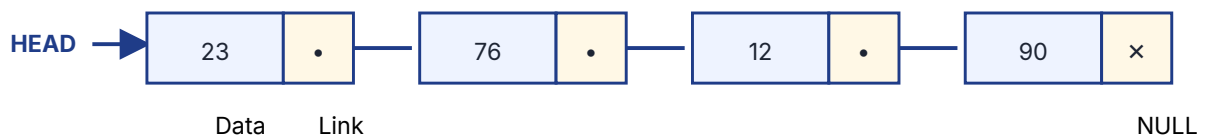
A **linked list** stores data as a chain. Unlike an array, the items are **not** one continuous block. Each separate piece is a **node** sitting somewhere in memory.

Each node has two parts:

1. the actual **data**;
2. a **link** (pointer) holding the memory address of the **next** node.

The first node is pointed to by **HEAD**. The last node's link is **NULL** (nowhere). To find an item you start at HEAD and follow links one by one — **sequential access**, $O(n)$.

Real-life example: browser history. Clicking Back follows the pointer to the previous page-node.



Follow HEAD → 23 → 76 → 12 → 90 → NULL

Four nodes. The last link is NULL, which ends the chain.

WORKED EXAMPLE

Worked example: walk from HEAD to print every data field.

```

nodes = [23, 76, 12, 90]
for data in nodes:
    print(data, end=" → ")
print("NULL")
  
```

OUTPUT

23 → 76 → 12 → 90 → NULL

There is no `nodes[2]` jump in a real linked list. The loop above models following each link once. Inserting a new node between 76 and 12 only means changing two pointers — you do not shift the rest of a big block, which is why lists beat arrays for frequent insert/delete.

REMEMBER IT

Array = row of lockers with numbers on the doors. Linked list = treasure hunt: each clue (link) tells you where the next box is.

COMMON MISTAKE

Mistake: writing `list[3]` and calling it linked-list access.

Fix: a true linked list has no index. Python's `list` type is an array-like structure. In this chapter, "linked list" means HEAD-and-pointers.

اردو میں سمجھیں

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CHECK YOURSELF

1. Name the two parts of a node.
2. What does the last node's link store?
3. Why is browser history a fair analogy?

Answers: 1. Data, and a link/pointer to the next node. 2. NULL (no next node). 3. Back follows the pointer to the previous page-node instead of jumping by index.

Next lecture: Linear Data Structures: Stack